improving the income of 100,000 plus farmers and generating more than 1 billion units of clean energy (kWh).

Ecozen's solutions reformed the irrigation and cold-chain industries with Ecotron (smart AI and IoT-assisted solar-pumping solution) and Ecofrost (smart AI and IoT-aided solar-powered cold room).

With Ecofrost, farmers are empowered to reduce food waste by 15-20%, increase earnings by up to 50% and allow their produce to reach markets valued at more than 10X. This is done by ensuring effective pre-cooling and cold storage of produce at the optimum temperature (4° and above) and relative humidity (65% to 95%) levels, with up to 30 hours of batteryless backup.

Ecotron provides farmers with stable and efficient irrigation, especially in areas with inconsistent or no electricity supply. It helps increase farm yield, reduces operating costs and supplements farmer earnings.

Global AI Mission in Agriculture

Overseas, several countries are clinging to digital tools to springboard the concept of modernized farming. "The AI treatment of agriculture is a tailwind to the whole global economy. According to the United Nations' prediction data on population and hunger, the world's population will increase by two billion within 2050, requiring a 60 per cent jump in food production to feed the total headcount. AI and ML can help narrow this gulf between food stock and the anticipated demand for sustenance to feed additional mouths," says Srijan Pal Singh, founder and CEO, Homi Lab. An IIM Ahmedabad alumnus and the co-author of the title Reignited With Dr. Kalam, Singh was the former advisor on policy and technology to Dr. A.P.J. Abdul Kalam, the 11th President of India.

Understanding these factors, a number of countries around the world took leaps in mingling AI with their agriculture, including developed nations like USA and the Western European countries like UK, Germany, France, Spain, among others. But there are other nations as well in this herd, especially those with huge population bases. While China and Singapore are leading the way in Asia, Iran has plans set for digital agriculture in the next couple of years. Closer home, India too is not lagging behind in giving a shot at sustainable farming.

Asia Pacific is the fastest-growing belt in the digital agriculture market. Skyrocketing modernization in the agriculture industry renders an impetus to this growth. The region is dominated by agriculture-dependent countries, such as India, China and Bangladesh. China already made significant investments in its agricultural industry using technologies like AI farming, sensors, drones and auto-steering to scale up efficiency and productivity.

Avant-Garde Techniques

Due to space crunch and the rise in pollution levels, experimentation finds its way through the avenues of agriculture. The science of hydroponics is adopted in many urban areas through terrace, balcony and tank farming on the surface of gravel, sand and liquid (mainly water) with added nutrients and without soil. Plus, vertical gardens are grown on the walls and building facades, while grassy tops are added to the flat roofs of moving cabs for fresh air and adequate green cover. Even veggies are being sprouted at the International Space Station farming lab.

The confluence of mathematical models and nature makes a perfect combo. After all, every plant depends on land, air, light, water and soil or any alternative nourishing medium for growth. Using these major variables and their sub-variables like heat, acidity level, water quality etc., multiple models

for predicting the plant traits can be made and managed. The hardware setup used in AI-driven hydroponic vertical gardens typically includes sensors that collect data on the plants' vital parameters, such as pH levels (acidity level) and nutrient supply. These sensors are placed near the plant roots and can also detect light density, temperature and humidity levels.

A visual camera can also be used to monitor the plants for any changes in coloration. The data collected by the sensors is then fed into an AI software tool, which uses machine-learning algorithms to analyze the info and lend insights into the plants' urgent needs.

AI can also prove to be advantageous to hybrid farming by paring up two ideal plants to produce the best traits in the resultant sapling. Machine learning as part of AI to be specific has been deft to date in identifying different zones in the plant's DNA. It can predict the location of genome crossovers. These are the regions where genetic material is exchanged between the paternal and maternal genomes. "By specifically targeting these areas, new hybrid varieties can be produced with the intended features. For instance, a recent study in Brazil showed that AI can create different varieties of sugarcane and forage, and even predict their performances. In terms of accuracy compared with habitual breeding techniques, the proposed methodology improved predictive powers by more than 50%," says Singh.

AI-powered agriculture is an exciting future for India, and it's going to be surreal to see drones and self driving tractors in our farmlands. What a sight that will be, right? d